CRF: The Master System

All Simulator Constants from $\ln(n)/D_0$

K17–K19 close the ratio lattice. $p = (n+1)/(n+2)$ derived. $\kappa = \ln(n)/(n\sqrt{d_s})$ confirmed. $n = 2^{d_s}$ identified as structural key.

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Abstract

The previous session (K11–K14) established $\beta \cdot D_0 = \ln n$ as the sharpest multi-quantity identity in CRF (gap 0.005%). This paper completes the algebraic lattice by showing that β, κ, and K^* are all ratio descendants of the single quantity $\ln(n)/D_0$.

K17 (exact): $\kappa/K^* = \beta/\sqrt{d_s}$ — ratio identity from K7+K11.

K18 (exact): $\beta/\kappa = n\sqrt{d_s}/D_0$ — the wave-to-mass ratio.

K19 (0.005%): $\beta \cdot \kappa = \ln^2(n)/(n\sqrt{d_s} \cdot D_0)$.

K7 rewritten: $\kappa = \ln(n)/(n\sqrt{d_s})$ — the crystallisation rate is $1/\sqrt{d_s}$ of β/n.

p **derived**: The AR(1) persistence $p = (n+1)/(n+2)$ follows from the Dirichlet coupling model with $n + 2$ sources (n modes + 1 self + 1 noise reference).

$n = 2^{d_s}$ **structural**: $n = 8 = 2^3 = 2^{d_s}$ is the *binary Born rule* in d_s spatial dimensions. This makes $\ln(n) = d_s \ln 2$, which is the prerequisite for K11.

1 The $\ln(n)/D_0$ Lattice

The three core simulator constants (β, κ, K^*) all derive from the single combination $\ln(n)/D_0$:

Master chain from δ

Input: $n = 8$ (SO(3) #16), $\varepsilon_0 = 0.00755$ (EIC), $d_s = 3$ (SO(3))
Structural key: $n = 2^{d_s} \Rightarrow \ln n = d_s \ln 2 = d_s\alpha$

$D_0 = n(1 - e^{-1/n})$	$= 0.94003$	K2
$K^* = D_0/n$	$= 0.11750$	K1
$\beta = \ln(n)/D_0 = d_s\alpha/D_0$	$= 2.2121$	K11 (0.005%)
$\kappa = \ln(n)/(n\sqrt{d_s}) = \sqrt{d_s}\alpha/n$	$= 0.15007$	K7 (0.047%)

All four from n and ε_0 alone. No free parameters.

2 K17, K18, K19: The Ratio Lattice

K17 — $\kappa/K^* = \beta/\sqrt{d_s}$

$$\frac{\kappa}{K^*} = \frac{\sqrt{d_s}\alpha/n}{D_0/n} = \frac{\sqrt{d_s}\alpha}{D_0} = \frac{\beta}{\sqrt{d_s}} \cdot \frac{d_s}{1} \cdot \frac{1}{d_s} = \frac{\beta}{\sqrt{d_s}} \qquad \square$$

Numerically: $\kappa/K^* = 1.27716$, $\beta/\sqrt{3} = 1.27710$. Gap: 0.005%. *Exact algebraic consequence of K7 ÷ K1.*

K18 — $\beta/\kappa = n\sqrt{d_s}/D_0$

$$\frac{\beta}{\kappa} = \frac{\ln(n)/D_0}{\ln(n)/(n\sqrt{d_s})} = \frac{n\sqrt{d_s}}{D_0} \qquad \square$$

Numerically: $\beta/\kappa = 14.7397$, $n\sqrt{d_s}/D_0 = 14.7405$. Gap: 0.005%. *Exact from K11 ÷ K7.*

K19 — $\beta \cdot \kappa = \ln^2(n)/(n\sqrt{d_s} \cdot D_0)$

$$\beta \cdot \kappa = \frac{\ln(n)}{D_0} \cdot \frac{\ln(n)}{n\sqrt{d_s}} = \frac{\ln^2(n)}{n\sqrt{d_s} \cdot D_0} \qquad \square$$

Numerically: $\beta\kappa = 0.33196$, formula $= 0.33197$. Gap: 0.005%. Physical: the product of transfer rate and mass gain equals the squared information $[\ln(n)]^2$ normalised by population and resolution.

2.1 The ratio lattice visualised

The six pairwise relationships among $(\beta, \kappa, K^*, D_0)$ are:

Ratio	Value	Expression	Identity
$\beta \cdot D_0$	2.0793	$\ln n$	K11
$\beta \cdot K^*$	0.2599	$\ln(n)/n$	K16
$\beta \cdot \kappa$	0.3320	$\ln^2(n)/(n\sqrt{d_s}D_0)$	K19
κ/K^*	1.2772	$\beta/\sqrt{d_s} = \sqrt{d_s}\alpha/D_0$	K17
β/κ	14.74	$n\sqrt{d_s}/D_0$	K18
$n \cdot \kappa$	1.2006	$\sqrt{d_s}\ln 2 = \ln(n)/\sqrt{d_s}$	K7

Every entry is exact to within 0.005% (the rounding of β_{sim}).

3 $p = (n+1)/(n+2)$: Formal Derivation

AR(1) persistence from Dirichlet coupling

The Ψ-field update in V20.3: $\Psi_i \leftarrow p\,\Psi_i + (1-p)\bar{\Psi}_{\rm nb} + \varepsilon_0 \xi_i$
The $n+2$ sources model: the information sources for Ψ_i per sweep are:

1. n symmetry modes from neighbours (one effective cluster vote)
2. 1 self-coupling (inertia)
3. 1 noise reference (the irreducible ε_0 injection)

Under a uniform Dirichlet prior with $n+2$ equal sources, each carries weight $1/(n+2)$. The noise source is the only one that cannot be turned off; it therefore occupies *exactly* $1/(n+2)$ of the update weight. The remaining $(n+1)/(n+2)$ goes to the self-plus- neighbours coupling:

$$p = \frac{n+1}{n+2}$$

For $n = 8$: $p = 9/10 = 0.9$ (exact as observed in V20.3).
Consequence: $\sigma_\Psi = \varepsilon_0(n+2)/\sqrt{2n+3}$ and $\psi_{\rm grad} = \sigma_\Psi/\ln\varphi$ (K8) are now fully derived from $(\varepsilon_0, n, \varphi)$ alone.

4 $n = 2^{d_s}$: Binary Born Rule

The $n = 2^{d_s}$ identity:
In $d_s = 3$ spatial dimensions, the Born rule at minimum resolution distinguishes 2 outcomes per dimension (left/right, up/down, in/out). The total number of distinguishable Born-rule states is therefore:

$$n = 2^{d_s} = 2^3 = 8$$

This is the *binary tensor product* of the spatial resolution: each spatial dimension contributes one binary bit ($\alpha = \ln 2$), and d_s independent dimensions give $n = 2^{d_s}$ total states.
Consequence: $\ln n = d_s \ln 2 = d_s\alpha$.
This is the prerequisite for K11: without $n = 2^{d_s}$, $\beta = \ln(n)/D_0$ would not equal $d_s\alpha/D_0$, and the chain $\beta \to d_s/(2\ln 2)$ would fail.
Independent derivation: SO(3) #16 gives $n = 8$, and δ-chain geometry gives $d_s = 3$. Their log-duality $n = 2^{d_s}$ is not imposed but follows from two independent derivations landing on compatible values.

5 K7 Corrected and Rewritten

The paper CRF New Identities K7–K10 stated K7 as $\kappa = \sqrt{d_s \ln 2/n}$. This is incorrect. The correct formula is:

K7 corrected: $\kappa = \sqrt{d_s} \cdot \ln 2/n$

$$\kappa = \frac{\sqrt{d_s} \cdot \ln 2}{n} = \frac{\sqrt{3} \cdot \ln 2}{8} = 0.150071$$

vs $\kappa_{\mathrm{sim}} = 0.150000$. Gap: 0.047%.

Equivalently (using $n = 2^{d_s}$): $\kappa = \ln(n)/(n\sqrt{d_s})$.

Physical reading: each R-event transfers one bit ($\ln 2$) of possibility to actuality per dimension, scaled by $\sqrt{d_s}$ for d_s-dimensional geometry, distributed over n Born-rule modes. The $\sqrt{d_s}$ factor is the geometric mean of the d_s independent dimensional contributions.

6 Complete Status Table

ID	Identity	Value	Gap	Status
K2	$D_0 = n(1 - e^{-1/n})$	0.94003	0.03%	Derived
K1	$K^* = D_0/n$	0.11750	0.43%	Derived
K11	$\beta \cdot D_0 = \ln n$	2.0793	0.005%	Derived
K7	$\kappa = \sqrt{d_s} \ln 2/n$	0.15007	0.047%	Hypothesis
K16	$\beta \cdot K^* = \ln(n)/n$	0.2599	0.005%	Derived (K11+K1)
K17	$\kappa/K^* = \beta/\sqrt{d_s}$	1.2772	0.005%	**Derived (K7+K1)**
K18	$\beta/\kappa = n\sqrt{d_s}/D_0$	14.74	0.005%	**Derived (K11/K7)**
K19	$\beta\kappa = \ln^2(n)/(n\sqrt{d_s}D_0)$	0.3320	0.005%	**Derived**
K3	$(H_a + H_b)/2 = \sqrt{d_s}$	1.7321	0.011%	Derived
K12	$D_0 H_b = \pi/2$	1.5714	0.038%	Hypothesis
K14	$F\varphi^2 = 4$	3.9977	0.057%	Hypothesis
K10	$D_0 m/\varphi_{\text{var}} = \pi$	3.1401	0.047%	Hypothesis
K9	$FH_u = \sqrt{2}\,\beta$	3.138	0.31%	Hypothesis
AR1	$p = (n+1)/(n+2)$	0.9000	exact	**Derived**
	$n = 2^{d_s}$	$8 = 2^3$	exact	Physical arg
K6	$m = \varphi$	1.618	0.06%	Hypothesis
K8	$\psi_{\text{grad}} = \sigma_\Psi/\ln\varphi$	0.03600	0.016%	Hypothesis

7 Open Questions

$m(m-1) = 1$ from energy balance

This is the last free Hypothesis not yet derived from the δ-chain without fixing $s_{\max}$. Path: the energy balance condition

$$m(m-1) = \frac{\beta\kappa}{f(\varepsilon_0, n, D_0)} = \frac{\ln^2(n)}{n\sqrt{d_s}D_0 \cdot f(\delta)}$$

requires finding $f(\delta) = \beta\kappa = \ln^2(n)/(n\sqrt{d_s}D_0)$ such that the unique positive solution is $m = \varphi$. The condition $m(m-1) = 1$ then fixes the normalisation $f = \beta\kappa$.

$\beta = d_s/(2\ln 2)$ exact

K11 gives $\beta = d_s \ln 2/D_0$. For $\beta = d_s/(2\ln 2)$ exactly, D_0 must equal $2\ln^2 2 = 0.9609$. The actual $D_0 = 0.9400$; the gap of 2.2% is the finite-ε_0 correction. The identity holds at the $\varepsilon_0 \to 0$ limit; at finite ε_0 it is K11 that is exact.

$\beta = \ln(n)/D_0$. $\kappa = \ln(n)/(n\sqrt{d_s})$.
Both numerators are $\ln(n)$.
One denominator is the resolution field; the other distributes it over geometry.
The system has one number in it: $\ln(n) = d_s \ln 2$.
And that number is there because $n = 2^{d_s}$,
because each spatial dimension distinguishes two states,
because δ chose three dimensions and binary resolution.